

Separate Pure Math Problems and Their Solutions

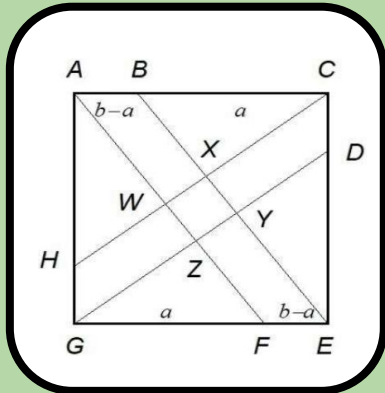
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Introduction: Solving problems in open math can be exploratory and interesting, but require dedication and patience, as well as creative thinking when arriving at a solution. This is especially true when approaching problems from two very unique fields of mathematics: geometry and set theory. While both questions appear short and straightforward, solving each of these problems requires much more of a time commitment and abstract thinking that one might expect.

Problem: J-5 from MathAMATYC Educator due 25 January 2024:

Proposed by Stephen L. Plett and Michael W. Ecker (jointly). ACEG is a square with sides of length b . Divide each side into segments of length $b-a$ and a and connect said segments as shown. Prove that WXYZ is a square, and then find the area of WXYZ as a fraction of the area of ACEG.



It is given from Figure 1 that the outer angles $\angle GAC$, $\angle ACE$, $\angle CEG$, and $\angle EGA$ are 90° and that the 2 line segments $\overline{AF}$ and $\overline{BE}$ are parallel as well. Referring to the named angles on the figure, α and β are created by parallel lines which means they are complimentary angles. It is also clear that $\delta = \beta$. Since δ and α are also complimentary, therefore γ is 90° . From the assertions above, it is found that lines $\overline{AF}$ and $\overline{HC}$ are perpendicular lines which means that parallelograms ABEF and DCHG, which are identical, intersect perpendicularly. This shows that the polygon in the center is a square.

The area of WXYZ as a fraction of ACEG can be found by separating the area of ACEG which is b^2 into the addition of the polygons AFG, ABEF, and BCE as shown in Figure 2. The area of the parallelogram ABEF can be represented by $L \cdot H$ where $H = \sqrt{a^2 + b^2}$ and the area of the 2 triangles together can be represented by ab . Solving for L in $b^2 = L\sqrt{a^2 + b^2} + ab$ where L represents a side length of WXYZ, it is found that $L = (b^2 - ab)/\sqrt{b^2 + a^2}$. The area of the WXYZ as a fraction of ACEG solved by squaring L and simplifying which results in $L^2 = b^2 \frac{(b-a)^2}{(a^2+b^2)} = b^2 \frac{(a-b)^2}{(a^2+b^2)}$.

Problem 2: J-5 from MathAMATYC Educator due 25 January 2024:

Proposed by Albert Natian. Consider the set of all permutations of the natural numbers. That is,

$$S = \{p : \mathbb{N} \rightarrow \mathbb{N}, p \text{ a permutation}\},$$

with $\mathbb{N} = \{\text{natural numbers}\}$. Is S countable? Proof?

Suppose S is countable. Then we can list out every possible permutation of $\mathbb{N}$. We do this by the form of an $\aleph_0 \times \aleph_0$ matrix, where each row represents a permutation of $\mathbb{N}$. Yet, we arrange the list of permutations in such a way where the diagonal elements of the matrix are unique from one another (e.g. the diagonal itself will form a permutation of $\mathbb{N}$).

Let c denote the cardinality of the continuum, which is equivalent to $2^{\aleph_0}$. By Cantor's Theorem, the powerset of $\mathbb{N}$ has strictly greater cardinality than $\mathbb{N}$, and as such $c > \aleph_0$. By the Schröder-Bernstein Theorem, if there are injections from one set to another, and vice versa, then the cardinality of those sets are the same. Therefore, the set of arbitrary functions, denoted A , consisting of functions, say $q : \mathbb{N} \rightarrow 2^{\aleph_0}$ has cardinality c . The set S , in this case, is a subset of this set A , and therefore also has cardinality c . By Cantor's diagonal argument, we can construct a permutation from the diagonal of this matrix such that it is not of the ones listed in the matrix rows, deeming the set S , contradictory to our hypothesis, uncountable. $\square$

